

Internship Machine Learning for Neuroscience



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Keywords. Brain imaging, causal discovery, representation learning, Python.

Targeted departments. Electrical & Computer Eng., Computer Science.

Expected outputs. Reproducible data-processing pipeline, MNE-Python tutorial.

Context. Progress in artificial intelligence has been driven in large part by extensive benchmark datasets. These datasets allow researchers to develop algorithms, compare them under common conditions, and build on one another’s results. Applying this approach to neuroscience has historically been difficult: collecting brain recordings is time-consuming, expensive, and technically demanding.

In recent years, however, large datasets of non-invasive brain recordings—including EEG, MEG, and fMRI—have become publicly available. At the same time, open-source Python libraries have made these data increasingly accessible to machine-learning researchers. Our recent work has drawn on these resources to learn representations [Banville et al., 2021, Heurtebise et al., 2025] and causal structure [Heurtebise et al., 2026] from brain-imaging data.

Goals. The goal of this project is to develop the computational infrastructure needed to apply machine-learning methods to brain recordings. The candidate will learn to store, organize, process, and analyze large neuroscience datasets on our group’s computing server.

The internship is particularly well suited to graduate students in machine learning who want to gain experience with a new data modality. Over six months, the candidate will build a foundation for future work in AI for science, particularly AI for neuroscience.

The project will proceed through three main stages:

- **Download and organize neuroscience datasets.** The candidate will set up datasets such as CamCAN [Shafto et al., 2014] and NeuralBench [Banville et al., 2026] on the group’s server and document the resulting data organization.

AI tools may be used and are encouraged when appropriate. However, the candidate must understand the resulting code and be able to explain and maintain it independently.

- **Learn MNE-Python.** MNE-Python is a widely used library for processing EEG and MEG recordings. The candidate will work through selected [MNE-Python tutorials](#) to become familiar with its core tools and data structures.
- **Develop reproducible analysis pipelines.** The candidate will use MNE-Python to process and analyze the downloaded datasets. This will consist in reproducing existing pipelines for learning causal structure and representations from brain recordings, such as [LiMVAM](#), as well as developing new pipelines.

Expected outcomes. This is a code-first project requiring curiosity and independence. The

candidate will also learn the relevant neuroscience concepts through selected readings and tutorials.

By the end of the internship, the candidate will be familiar with modern computational tools for machine learning on brain recordings, including dataset organization, signal processing, and the reproduction of recent research pipelines. This experience will provide a strong foundation for conducting and publishing research in machine learning for neuroscience.

The candidate's work will be documented and acknowledged in the group's GitHub repository. They will also deliver a tutorial presenting the pipeline to graduate students and faculty.

Environment. The position is based in my [research group](#) at the University of Wisconsin–Madison. We work on machine learning, with a particular focus on generative models. Our research combines mathematical foundations, practical algorithm development, and applications in areas such as brain imaging and molecular science.



Requirements. We are seeking a highly motivated master's or early-stage PhD student with strong programming skills and an interest in applying machine-learning methods to brain data. Exceptional undergraduate students with a strong technical background may also be considered.

Experience with Python is required. Particular attention will be given to the candidate's ability to write clean, clear, and well-documented code. Prior experience with neuroscience or brain-imaging data is welcome but not required — curiosity is.

How to apply. Please email ochehab@wisc.edu with the subject line [Internship] AI for Brain Imaging and attach:

1. a CV named `firstname_lastname_CV.pdf`, including a link to your Google Scholar profile, if applicable;
2. a short statement named `firstname_lastname_cover.pdf`, containing one paragraph explaining your interest in the position and what you would bring to the project.

References

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